

EFY Note

The source code of this project is available for free download at source.efymag.com

made high, BC547 transistor T4 conducts, relay RL4 is energised and load connected across RL4 turns on.

Software program

Main operation is done by the software program loaded into the internal memory of Arduino Nano. The program is written in Arduino programming language or sketch. Arduino IDE 1.6.8 is used to compile and upload the program.

ATmega328P on Arduino Nano board comes with a pre-programmed bootloader that allows you to upload a new code to it without using an external hardware programmer.

Connect Arduino Nano board to the PC and select the correct COM port in Arduino IDE. Compile the program/sketch (joystick_control.ino) and select the correct board from Tools→Board menu in Arduino IDE. Upload the sketch to load the program to the internal memory of the MCU. The sketch is at the heart of the system and carries out all major functions. In this project, external header files are not required for programming. It is a simple way to detect joystick values, and

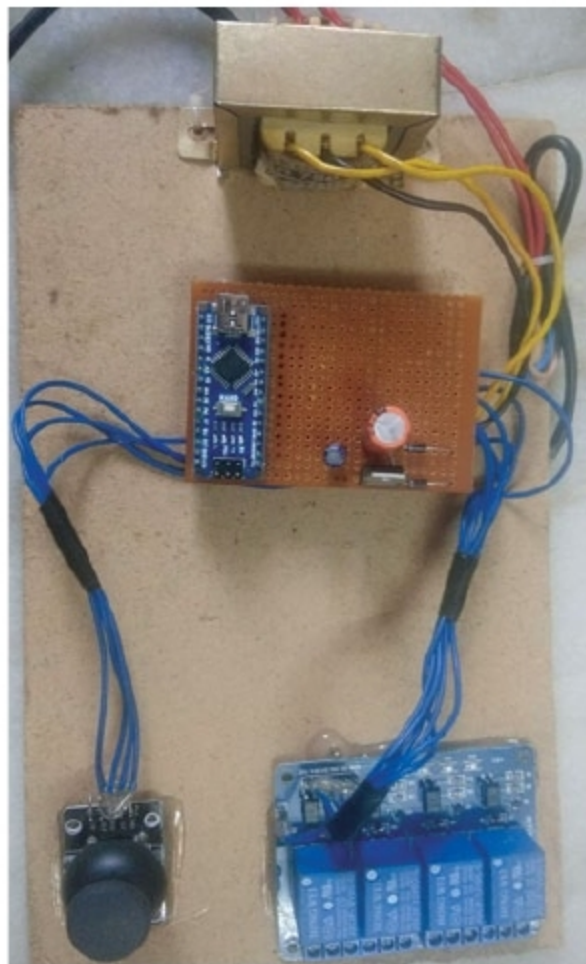


Fig. 6: Author's prototype

compare VRx and VRy values using if-else ladder functions to control four appliances.

Construction and testing

An actual-size PCB layout of the joystick-controlled industrial automation system is shown in Fig. 4 and its components layout in Fig. 5. Connect the power supply from the transformer to the PCB at X1. Connect joystick output wires to the PCB at JM1. After loading the program into Arduino Nano, as described earlier, mount it on the PCB on the space provided for it. Switch on the power supply. Now you can move the shaft left, right, up and down to control the electrical appliances. The author's prototype is shown in Fig. 6. **EFY**



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